

Haoting Alexa Yu

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EDUCATION

University of British Columbia

BSc in Mathematics, Dean's Honor List

Vancouver, Canada

Sep 2018 - May 2022

University of Edinburgh

MSc in Operational Research with Data Science, Merit

Edinburgh, UK

Sep 2022 - Sep 2023

University of the Arts London

MRes in Creative Computing

London, UK

Sep 2025 - Sep 2026

PUBLICATIONS

From Retrieval to Responsibility: Designing Creative MRAG with Control and Governance

- **Haoting Yu**, Zhiyan Jiang, Elizabeth Wilson, Yunrui Yang. In *Proceedings of the 2026 Conference on Creativity and Cognition (C&C '26)*, 1478–1483. ACM. DOI: 10.1145/3803784.3816842
- Designed a three-axis control and governance framework for creative MRAG; interviewed 10 visual and sonic creators and derived design implications for provenance, interaction, and responsible reference-based generation.

Who Chooses the Artwork? Curatorial Agency and Distributed Intent in Botto

- **Haoting Alexa Yu**, Bea Wohl. In *Proceedings of the Fourth International Workshop on eXplainable AI for the Arts (XAIxArts 4)*, co-located with ACM C&C 2026.
- Analyzed Botto as an agentic curatorial system, showing how authorship is distributed across generation, model filtering, DAO voting, feedback, and recognition

RESEARCH PROJECTS

Personalized Poem Generation with Fine-Tuned Large Language Model

Jan 2025 – present

- Fine-tuned 7B LLM instruct model on 100 self-written poems using LoRA fine-tuning method to generate poetry that replicates my style, tone, and imagery, with title-based input for seamless poem generation

Towards Robust Automatic Drum Transcription via Domain-Randomized Data Synthesis

Mar 2026 – May 2026

- Benchmarked the ADTOF automatic drum transcription model on a synthetic drum dataset; prepared drum-event annotations, ran baseline inference and evaluation, and reviewed prior ADT literature to inform baseline selection and interpret model performance.

Interactive Semantic Mapping of Multilingual Free-Text Responses

Sep 2025 – Dec 2025

- Developed an interactive web-based artwork mapping anonymous multilingual free-text responses into a navigable semantic star field using text embeddings, UMAP, KMeans clustering, and cosine-similarity links; examined how machine-derived similarity aligns with or flattens human interpretations of emotional connection and belonging.

SKILLS

Language: Mandarin (native speaker), English(fluent)

Programming: Python, R, SQL, MATLAB, C++ | **ML/AI:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, SciPy; multimodal AI, RAG, LLM fine-tuning, generative AI

SELECTED WORK & TEACHING EXPERIENCE

Fugro

Wallingford, UK

Laboratory Quality Engineer

Aug 2024 – Mar 2025

- Automated geotechnical data processing in Python to improve efficiency, accuracy, and reproducibility; developed a YOLOv9 baseline for soil particle and grain detection.
- Initiated an R-based inter-lab comparison framework to evaluate and benchmark laboratory performance across locations.

HD Education

Remote

Mathematics Science Instructor

May 2022 - Present

- Provide one-to-one and group instruction in undergraduate calculus, algebra, and statistics, with tailored support for mathematical reasoning and problem solving.

UBC Faculty of Science

Vancouver, Canada

Math Department Teaching Assistant

Sep 2021 - May 2022

- Prepared course contents and weekly lab assignments
- Assisted students individually or in small groups with their assignments and related questions in facility workshop (teaching duty)